ChatGPT

Seal Swell After Aromatic Saturation in Hydrotreated Jet Fuel

Direct answer

If your hydrotreating step saturates a large fraction of the aromatics, the most defensible expectation is that **seal swell decreases**, even if you assume those aromatics are converted mainly into cycloparaffins rather than into straight or branched paraffins. The literature is fairly consistent on this point for nitrile-based aviation seals: aromatics are the strongest swell drivers, cycloaromatics and diaromatics are stronger still, cycloalkanes help somewhat, and normal/iso-alkanes are weak contributors. In other words, **cycloparaffins partially recover seal swell, but they do not fully replace the swell activity of the aromatics they came from**. [1]

A useful way to think about your hydrotreated jet fuel is that hydrotreating changes both the **chemical family balance** and the **interaction strength** between fuel and elastomer. FAA/Boeing's fuel-materials work found that fuel/elastomer interaction strength generally rises as fuel molecules become smaller, more polar, and more capable of hydrogen bonding; alkylbenzenes and especially diaromatics are therefore much more active than alkanes, while cycloalkanes sit in between. The same FAA study also showed that nitrile O-rings are among the most composition-sensitive materials, whereas fluorosilicones are much less dependent on aromatic content. [2]

So for a hydrotreated jet model, the right first-pass assumption is not "aromatics converted to cycloparaffins, therefore seal swell is unchanged." The better assumption is: **aromatic saturation lowers seal swell; cycloparaffins offset only part of that loss; the magnitude of the drop depends strongly on elastomer type and on which aromatic subclasses were removed.** [3]

What the evidence says about class effects

The clearest modern class-by-class data set is Faulhaber, Borland, and Boehm's 2023 nitrile-rubber study. They measured 56 solvents and fuels, including 39 neat molecules doped at 8 vol% into a very low-swelling SAF. In that data set, average converged nitrile swell at 8 vol% followed this order: **naphthalenes 12.7% v/v**, **cycloaromatics 7.7%**, **alkylbenzenes 4.8%**, **polycycloalkanes 1.5%**, and **monocycloalkanes 1.2%**. They explicitly note that n-/iso-alkanes were not tested as dopants because prior work had already shown they have an insignificant influence on O-ring swell. That ranking is the strongest direct evidence for your hydrotreated-fuel question. [4]

Those same 2023 data make the "aromatic-to-cycloparaffin" penalty very tangible. In their concentration-series experiments, **8 vol% n-butylcyclohexane** gave about **1.0%** converged swell, while **8 vol% n-propylbenzene** gave **4.9%**, **8 vol% tetralin** gave **7.1%**, and **8 vol% 2-ethylnaphthalene** gave **12.1%**. By inference, if hydrotreating moves your pool from alkylbenzenes, tetralins, and naphthalenes toward mono- and bicyclic saturates, your finished fuel should swell nitrile seals less, not equally. [5]

The FAA/Boeing CLEEN report adds an important nuance that improves a refinery model. It found that aromatics absorbed into tested O-rings and sealants were, on average, **4.75 times more soluble than alkanes**, but it also estimated that **alkanes still contributed roughly half of the total volume swell** because they make up most of the fuel. That means your regression should not rely on total aromatics alone. It should also include the saturated portion of the fuel, especially its average molecular size or boiling-range distribution. The same report emphasizes that as the alkane distribution shifts to smaller, lighter molecules, their contribution to swell increases. [6]

That is why "all aromatics become cycloparaffins" is still too simple. You need at least three layers of information: **which aromatic subclasses were lost**, **what cycloalkane subclasses replaced them**, and **how the rest of the paraffinic pool shifted**. FAA/Boeing's analysis and Faulhaber's data both point in that direction, and Faulhaber further reports that across hydrocarbon classes, NBR seal swell is more linear with **density** than with **molar volume**, making density a particularly practical whole-fuel predictor to combine with class composition. [7]

The literature also shows that there is **no universal aromatic-content threshold that guarantees acceptable swell for every material**. A common heuristic in aviation fuel work is around **8 vol% aromatics**, and recent papers still reference that level as the conventional minimum surrogate for elastomer compatibility. But the FAA/Boeing statistical study found that, for one nitrile material, a 50/50 SPK/Jet-A blend constrained to **8% aromatics** had only **23% overlap** with the reference Jet-A swell range; even extracted nitrile reached only **34% overlap**. In other words, **8% aromatics is not a magic guarantee**, especially once chemistry shifts among aromatic subclasses or toward low-polarity saturates. [8]

Finally, real hardware behavior remains material-specific. Hamilton and coauthors, testing several nitriles and one fluorosilicone, found that Jet A produced appreciable nitrile swell, while **HEFA and ATJ-SPK tended to cause very small changes or shrinkage**, which they attributed to the absence of aromatics. In the same paper, different nitrile O-rings from the same broad family still behaved differently, reinforcing that your model should be **elastomer-specific**, not fuel-only. [9]

Papers worth reading first

If you want to build a regression instead of only a heuristic, the single best starting point is **Faulhaber, Borland, and Boehm, 2023, *Measurements of Nitrile Rubber Absorption of Hydrocarbons: Trends for Sustainable Aviation Fuel Compatibility***. It is the richest accessible hydrocarbon-class data set I found for jet-fuel-relevant species, and it gives you both class averages and concentration-series data for representative compounds. It also establishes a conventional-fuel reference range for the nitrile batch used in that work, which is useful for framing model outputs against an operationally familiar target window. [4]

The second paper to read is **Kosir, Heyne, and Graham, 2020, *A Machine Learning Framework for Drop-in Volume Swell Characteristics of Sustainable Aviation Fuel***. It trained a neural-network framework on **230 observations** across **10 non-metallic materials**, using neat-compound extrapolations from swell data. For nitrile rubber, extracted nitrile, and polythioether, the holdout errors were reported as **12.4% or better relative to mean volume swell**, and the work explicitly frames cycloalkanes as replacements for aromatics in high-performance jet fuel design. Even if you prefer a transparent regression over a neural net, this paper is the right performance benchmark. [10]

The third essential source is the **FAA/Boeing CLEEN report, *Impact of Alternative Jet Fuel and Fuel Blends on Non-Metallic Materials Used in Commercial Aircraft Fuel Systems***. This is the most useful source for turning a chemistry-only model into a compatibility model. It provides material-dependent sensitivity, “specific swell” slopes versus aromatic content, fuel/polymer partition-coefficient reasoning, statistical comparison to reference Jet-A populations, and explicit evidence that nitrile is much more aromatic-sensitive than fluorosilicone or fluorocarbon. For model design, it is especially valuable because it warns against using aromatic content alone. [11]

For exact molecule-structure effects, read **Romanczyk et al., 2019, *The Capability of Organic Compounds to Swell Acrylonitrile Butadiene O-rings and Their Effects on O-ring Mechanical Properties***. This paper is excellent for feature engineering. It shows that **ethylbenzene** and **indane** were especially effective swell promoters among the compounds tested, that shorter and less-branched alkylbenzenes tend to swell more strongly, that naphthene-containing aromatics can outperform same-carbon-number alkylbenzenes, and that saturated hydrocarbons such as cyclohexane cause little swelling. It also links swell to tensile-strength reduction, which is useful if you eventually want a multivariate compatibility model instead of a volume-only model. [12]

For compressed-seal behavior closer to engine hardware, read **Anuar et al., 2021, *Effect of Fuels, Aromatics and Preparation Methods on Seal Swell***. Unlike pure soak studies, this paper uses stress relaxation under compression at 30 °C for about 90 hours and shows that Jet-A1 swells more than SPK, that higher aromatic percentage increases swelling, and that aromatic identity matters. If your end use is practical compatibility rather than just a screening model, this paper helps you decide whether to predict only free-swell volume or to add a second model for compressed-seal response. [13]

Two additional papers are worth keeping nearby as context. **Graham et al., 2006, *Swelling of Nitrile Rubber by Selected Aromatics Blended in a Synthetic Jet Fuel*** is the classic early species-selection paper that later reviews and ML efforts repeatedly build on. And **Boehm et al., 2024, *Perspective on Fuel Property Blending Rules for Design and Qualification of Aviation Fuels*** is the best current review of how to combine regression models with blending rules; it reports that a **linear volumetric blending rule** for O-ring swell achieved an **MAE of 1.23% v/v** across **52 mixtures** and explicitly recommends building single-component swell models from molecular properties such as Hansen solubility parameters and structural descriptors. [14]

If your model will be used in an ASTM qualification workflow, keep **ASTM D4054** in view. ASTM describes D4054 as the practice for evaluating new aviation turbine fuels and additives, and the D4054 Clearinghouse guidance explains how candidate fuels move toward D7566 and then, if acceptable, into D1655-equivalent service. A regression model can be very useful for screening and formulation, but it does **not** replace fit-for-purpose testing. [15]

A regression strategy that fits the literature

The literature supports a **hybrid modeling strategy**, not a single-variable correlation with total aromatics. The best baseline is a **linear volumetric blending rule** for the finished fuel, because that rule has already been shown to work reasonably well for O-ring swell across measured binary and ternary mixtures. On top of that blending layer, your single-component or hydrocarbon-class coefficients should come from a regression model trained on class composition and molecular descriptors. This is almost exactly the path recommended by the 2024 blending-rules review. [16]

For the response variable, I would model **converged nitrile volume swell, % v/v**, first. That is the most data-rich target in the available aviation literature, and nitrile is also the material that shows the strongest dependence on aromatic removal. If you expect the fuel to contact multiple material families, build separate models for **nitrile**, **fluorosilicone**, and **sealants** rather than a pooled model, because the FAA/Boeing and Hamilton results both show materially different sensitivities. [17]

For predictors, the minimum defensible set is the following: **alkylbenzene vol%**, **cycloaromatic vol%**, **diaromatic or naphthalene vol%**, **monocycloalkane vol%**, **polycycloalkane vol%**, **n-/iso-alkane vol%**, and at least one whole-fuel descriptor such as **density** or a boiling-range proxy. If you have GC×GC or detailed hydrocarbon analysis, a better feature set is class composition plus **class-weighted Hansen solubility parameters**, **average molar volume**, and **average carbon number**. That recommendation follows directly from FAA's interaction framework, Faulhaber's density result, and the 2024 review's suggestion to model unmeasured compounds using observed relationships with molecular properties and Hansen parameters. [18]

A practical first regression form for nitrile would be:

$$S_{\text{NBR}} = \beta_0 + \beta_1 A_{\text{alkylbenzene}} + \beta_2 A_{\text{cycloaromatic}} + \beta_3 A_{\text{naphthalene}} + \beta_4 C_{\text{mono-cyclo}} + \beta_5 C_{\text{poly-cyclo}} + \beta_6 P_{\text{n/iso-alkane}} + \beta_7 \rho + \beta_8$$

where the expected ordering for nitrile, based on the published class data, is **β3 > β2 > β1 >> β5 ≈ β4 > β6**. That coefficient ordering is an inference from the available measurements, especially the 8 vol% dopant results in Faulhaber and the negligible n-/iso-alkane effects discussed there. [19]

I would then add a **residual correction model** rather than forcing all complexity into the first regression. A simple sequence is: first predict class-based swell; next apply the **linear volumetric blending rule** to get finished-fuel swell; then fit the residual error against descriptors that represent what the class model misses, such as **average alkylbenzene branching**, **share of fused bicyclic saturates**, or **trace heteroatom content**. That last term matters because the FAA/Boeing report notes that oxygenates, sulfur-containing species, and acidic compounds can also affect swell, and the review paper identifies elastomer compatibility as an impurity-sensitive property with sparse data. [20]

Validation matters as much as the equation form. I would not random-split these data. Instead, use **leave-one-fuel-out** or **leave-one-chemistry-family-out** validation, because you care about extrapolating from known fuel families to a newly hydrotreated pool. If you mix data sets, also validate by **elastomer batch** or at least by **study**, since the 2024 review notes that O-ring swell tests can vary between manufacturing batches and take days to weeks to complete. [21]

Practical assumptions for your hydrotreated jet model

For your specific refinery assumption, the right prior is this: **saturating alkylbenzenes into alkylcycloalkanes reduces swell sharply; saturating cycloaromatics and naphthalenes into fused bicyclic saturates also reduces swell, but the remaining cycloalkanes still provide more swell than an all-paraffinic pool would.** That statement is not a direct one-to-one reaction experiment, but it is the inference most consistent with the published class averages and single-species concentration data. [22]

What I would not do is build the model around only **total aromatics wt%**. The FAA/Boeing study shows why: aromatics are much more soluble in the polymers than alkanes, but the alkanes still contribute about half the total swell because they dominate the fuel; meanwhile, different materials respond differently, and some low-aromatic blends can still fall outside reference nitrile behavior even near the familiar 8% aromatic level. A model with only total aromatics will therefore miss both the structural chemistry of the aromatic fraction and the changing role of the saturated pool after hydrotreatment. [23]

A better refinery-side shortcut is to use a **seal-swell potential index** whose weights are initialized from the literature and then calibrated to your own test data. For nitrile, the literature suggests the weights should strongly favor **naphthalenes**, then **cycloaromatics**, then **alkylbenzenes**, with **polycycloalkanes** and **monocycloalkanes** much smaller but still nonzero, and **n-/iso-alkanes** smallest of all. The exact coefficients should come from your plant's fuels and your target seal material, but that ordering is supported by the measured class averages and by the broader FAA interaction framework. [24]

If your purpose is formulation rather than post hoc prediction, the same literature points to a strategy. Small, lightly substituted aromatics can provide swell with less soot penalty than heavy polyaromatics, and DOE's SAF review notes that fused-ring cycloalkanes may offer similar seal-swelling functionality with better combustion behavior than aromatics. That means a formulation model should not ask only "how many aromatics remain after hydrotreating?" but also "which aromatics should be preserved, and which cycloalkanes are best at backfilling compatibility?" [25]

The short version is that **hydrotreating a jet fuel by saturating aromatics into cycloparaffins should lower seal swell, often materially for nitrile-based systems**. For developing a regression, start with **Faulhaber 2023**, **Kosir 2020**, the **FAA/Boeing CLEEN report**, **Romanczyk 2019**, and **Anuar 2021**, then build an **elastomer-specific hybrid model** that combines hydrocarbon classes, density or volatility proxies, and a linear mixture rule. That approach aligns most closely with the best primary sources currently available. [26]

---

[1] [3] [4] [5] [19] [22] [24] [26] Measurements of Nitrile Rubber Absorption of Hydrocarbons: Trends for Sustainable Aviation Fuel Compatibility
https://rosap.ntl.bts.gov/view/dot/68095/dot_68095_DS1.pdf

[2] [6] [7] [11] [17] [18] [20] [23] Impact_of_Alternative_Jet_Fuel_and_Fuel_Blends_on_Non-Metallic_Materials_Used_in_Commerical_Aircraft_Fuel_Systems_CLEEN
https://www.faa.gov/sites/faa.gov/files/about/office_org/headquarters_offices/apl/Impact_of_Alternative_Jet_Fuel_and_Fuel_Blends.pdf

[8] Measurements of Nitrile Rubber Absorption of Hydrocarbons
https://pubs.acs.org/doi/10.1021/acs.energyfuels.3c00781?utm_source=chatgpt.com

[9] cambridge.org
https://www.cambridge.org/core/services/aop-cambridge-core/content/view/0863C07931DFA895AC87381992D47778/S0001924024000599a.pdf/div-class-title-investigation-on-elastomer-behaviour-when-exposed-to-conventional-and-sustainable-aviation-fuels-div.pdf

[10] A machine learning framework for drop-in volume swell characteristics of sustainable aviation fuel
https://rosap.ntl.bts.gov/view/dot/55599/dot_55599_DS1.pdf

[12] The capability of organic compounds to swell acrylonitrile butadiene O-rings and their effects on O-ring mechanical properties
https://www.triceceramics.com/uploads/9/3/8/9/93899770/49_oring_article_2019.pdf

[13] Effect of fuels, aromatics and preparation methods on seal swell | The Aeronautical Journal | Cambridge Core
https://www.cambridge.org/core/journals/aeronautical-journal/article/effect-of-fuels-aromatics-and-preparation-methods-on-seal-swell/CF929FF16F530C8ECBB1D454872E9EA4

[14] A Statistical Approach to Estimating the Compatibility of ...
https://www.researchgate.net/publication/268482439_A_Statistical_Approach_to_Estimating_the_Compatibility_of_Alternative_Fuels_with_Materials_used_in_Seals_and_Sealants?utm_source=chatgpt.com

[15] D4054 Standard Practice for Evaluation of New Aviation ...
https://www.astm.org/d4054-22.html?utm_source=chatgpt.com

[16] [21] Perspective on Fuel Property Blending Rules for Design and Qualification of Aviation Fuels: A Review
https://rosap.ntl.bts.gov/view/dot/79575/dot_79575_DS1.pdf

[25] Sustainable Aviation Fuel: Review of Technical Pathways Report
https://www.energy.gov/sites/prod/files/2020/09/f78/beto-sust-aviation-fuel-sep-2020.pdf